

README_v1.5_FINAL_FREEZE.md

MonitorAI

Version 1.5 FINAL FREEZE

Status: Approved For Development

Overview

MonitorAI is an iPhone-based orthodontic image acquisition platform designed for:

- Remote orthodontic monitoring
- Standardized image collection
- Future AI dataset generation

MonitorAI is not a diagnostic AI.

The primary objective is to collect highly standardized clinical images and metadata suitable for future machine learning applications.

Core Mission

Create the highest-quality orthodontic dataset possible while maintaining a workflow that patients can complete independently.

Every design decision should improve:

- Standardization
 - Repeatability
 - Clinical usefulness
 - Metadata quality
 - Future AI value
-

Target Environment

Client

iPhone Application

SwiftUI

Swift 6

MVVM

Supported Devices

iPhone 11 Series

iPhone 12 Series

iPhone 13 Series

iPhone 14 Series

iPhone 15 Series

iPhone 16 Series

iPhone 17 Series

Including:

- Standard
 - Plus
 - Pro
 - Pro Max
 - Air
-

Backend

FileMaker Server 19

FileMaker Data API

External Secure Storage

Clinical Workflow

Position Calibration



11-Step Capture Workflow



Review



Upload



Completion

Capture Protocol

Group A

Aligner ON

5 mm Open

Step01 Front ON

Step02 Right Buccal ON

Step03 Left Buccal ON

Group B

Aligner OFF

5 mm Open

Step04 Front OFF

Step05 Right Buccal OFF

Step06 Left Buccal OFF

Step07 Upper Occlusal

Step08 Lower Occlusal

Group C

Light Occlusion

Step09 Front Occlusion

Step10 Right Occlusion

Step11 Left Occlusion

Position Calibration

Purpose

Align retractor opening to screen center.

Requirements

Opening Center Error < 5%

Stable 3 Seconds

Automatic Advance

PTG

Progressive Target Guidance

PTG evaluates:

- Posterior visibility
- Occlusal visibility

- Clinical usefulness

PTG does not primarily evaluate head rotation angle.

Levels

Dot1–Dot6

Capture threshold:

Dot4

Auto Capture

Rule

$PTG \geq \text{Dot4}$

AND

Stable 4 Seconds

↓

Capture

Plateau Capture

Rule

$PTG \geq \text{Dot4}$

AND

No visibility improvement

AND

Stable 4 Seconds

↓

Capture

Purpose

Reduce excessive retakes.

Overlay System

Overlay Type

PNG

Transparent Background

Assets

Front_ON

Right_Buccal_ON

Left_Buccal_ON

Front_OFF

Right_Buccal_OFF

Left_Buccal_OFF

Upper_Occlusal

Lower_Occlusal

Front_Occlusion

Right_Occlusion

Left_Occlusion

Architecture

Pattern

MVVM

Layers

App

↓

Feature

↓

ViewModel

↓

Service

↓

Repository

↓

FileMaker Data API

↓

FileMaker Server 19

Core Services

CameraService

PositionCalibrationService

OverlayGuideService

PTGService

VisibilityScoringService

AutoCaptureService

PlateauDetectionService

VoiceGuidanceService

HapticFeedbackService

MetadataService

StorageService

UploadService

Image Requirements

Stored Image

Always 1.0x

Preview Zoom

2.5x

3.0x

4.0x

Preview only

Torch

Default 70%

Adaptive control allowed

Metadata Philosophy

Metadata is equal in importance to image data.

Every accepted image must store:

- PTG Data
- Visibility Data
- Focus Data
- Blur Data
- Exposure Data
- Calibration Data
- Timing Data
- Device Data

Dataset Integrity Fields

Required

ImageWidth

ImageHeight

ImageSizeBytes

ImageSHA256

Purpose

- Duplicate detection
- Integrity validation
- Future AI export

FileMaker Database

Core Tables

Patients

MonitoringSessions

CaptureImages

CaptureMetadata

UploadQueue

TrainingLabels

SystemSettings

Upload Strategy

Recommended Order

Create Session

↓

Create Capture

↓

Create Metadata

↓

Upload Image

↓

Upload Thumbnail

↓

Update Session

Offline Support

Capture workflow must function without network connectivity.

Failed uploads are stored locally and retried automatically.

Retry Schedule

1 minute



5 minutes



15 minutes



1 hour



Manual Retry

Security

Required

HTTPS

TLS 1.2+

Bearer Token

Keychain Storage

Dedicated API Account

External Secure Storage

Testing Requirements

Required

Unit Tests

Integration Tests

Device Tests

Clinical Tests

Dataset Validation

Definition of Success

MonitorAI is successful when:

Patients can independently create clinically useful images.

Clinicians can reliably review those images.

The resulting dataset supports future AI development.

Final Principle

MonitorAI is not a camera application.

MonitorAI is a standardized orthodontic image acquisition platform and future AI dataset generation platform.

Every technical decision should prioritize:

- Dataset quality
- Metadata quality
- Clinical usefulness
- Long-term AI value